

Q&A Portal Website
DO PHUONG VU
StudenID: GCS220099
UNIVERSITY OF GREENWICH
Lecturer: Nguyen Quoc Khanh
August 21 2024

COMP1841 (2023/24)	Web Programming 1	Contribution: 100% of course
Module Leader: Matt Prichard	PHP/MySQL CRUD system	Deadline Date: Tuesday 21/08/2024
Learning Outcomes: 1 Develop a web application with relational database connectivity using a variety of technologies, specifically HTML5, PHP PDO and MYSQL 2 Discuss issues surrounding Internet and intranet technologies, such as browser dependency, accessibility, legal and security concerns. 3 Critically analyse and evaluate Internet related business problems.		

Table of Contents

1. Introduction.....	4
2. System development.....	4
3. Technologies Used.....	5
4. Legal, Social, and Ethical Considerations	6
<i>Legal and Ethical Considerations</i>	<i>6</i>
5. System Overview	7
Function Of user:	8
Function Of admin:	10
6. Testing and Validation	13
7. Evaluation, Conclusion, and Future Recommendations.....	16
Reference:	17
Appendix:	17

1. Introduction

The purpose of this project was to create a working prototype of a web-based platform, akin to a scaled-down version of Stack Overflow, where students could submit and handle coursework-related issues. MySQL was used for data storage and PHP PDO for database interface throughout system implementation. This study provides an in-depth analysis of the development process, covering design decisions, technologies employed, legal concerns, and a functional assessment of the system.

2. System development

The system includes several main pages:

Homepage: Provides an overview of all the modules inside these modules, each of which will include posts with details such as the title, author, posting date, content, and any related photos. Users can modify (edit) or remove (delete) their own postings here, and other users can respond to that question by leaving a comment (comment button). Regarding Admin, it will be a page where all posts and comments are managed, including details about the poster, module_id, questions and answers content, photos, and posting time. It will also include an area where admins may submit questions, which is akin to a user function.

Post Page: This feature enables students to submit their questions by utilizing a pre-made form and entering details like the question's title, content, module name, and image (if applicable), before publishing it.

User Page: Users can modify their account password by first entering their old password, then entering and confirming the new one on the User Page. Admin has the ability to add, edit, and remove personal data, as well as transition between user and admin positions.

Module Management Page: Allows administrators to manage the names of deleted or added new modules.

Contact Form: Allows students to send messages by entering a pre-made form and then submitting it to the administrator by email.

Because the Bootstrap framework is used to provide a responsive layout and consistent navigation, every page is created with user accessibility in mind. With their clear and simple visual style, the form fields and buttons are easy to recognize.

The navigation structure of the Q&A portal features a main navigation bar with links to the Home Page, Post Management, User Management, Module Management, and a Contact Form. This layout provides a straightforward and user-friendly experience. Additionally, the system implements role-based access control, allowing administrators exclusive access to certain management functions, enhancing security and usability.

The relationships between posts, users, and modules are effectively managed by the underlying data structure, which is built with several interrelated tables. The primary tables are "Users," which hold vital user data like email addresses and usernames; "Posts," which house all student-generated content and have foreign keys connecting each post to the appropriate user and module; and "Modules," which hold the names of the different course modules that are offered on the platform. Foreign key constraints are used to preserve referential integrity, which keeps the database reliable and consistent by guaranteeing that each post is consistently linked to legitimate users and modules.

3. Technologies Used

PHP PDO

Because PHP Data Objects (PDO) offers security features including prepared statements that thwart SQL injection attacks, it was selected for database interaction. Additionally, PDO offers a consistent and adaptable database operation interface.

MySQL Information

Because of its dependability and compatibility with relational database systems, MySQL was chosen for data storing. Referential integrity and normalisation are prioritised in the database architecture to reduce duplication and guarantee data consistency.

CSS3 and HTML5

The web pages were styled using CSS3, and their structure was provided via HTML5. A responsive and aesthetically beautiful front end may be quickly developed thanks to the usage of the CSS framework Bootstrap.

Validation of JavaScript

JavaScript was used for client-side form validation, which minimised server load and gave users instant feedback. Aspects for HTML5 form validation were also included to improve user experience.

4. Legal, Social, and Ethical Considerations

It is imperative to take into account a range of legal, social, and ethical considerations when creating an online platform such as the Q&A site, with a particular focus on web accessibility and the management of personal data. In addition, adherence to the General Data Protection Regulation (GDPR) is essential to guaranteeing the legality of the platform and safeguarding user privacy.

Legal and Ethical Considerations

Web Accessibility:

Accessibility of websites is a crucial ethical and legal necessity. It is not only legally required by laws such as the Americans with Impairments Act (ADA) and the UK's Equality Act 2010 to make the platform accessible to all users, including those with impairments, but it is also morally right to allow equitable access to educational materials. To help users with visual impairments, elements like keyboard navigation, screen reader compatibility, alt text for images, and enough color contrast must be implemented. To satisfy these requirements, the platform must follow the Web Content Accessibility Guidelines (WCAG) 2.1.

Personal Data Storage:

Significant legal and ethical issues are brought up by the platform's handling of personal data, which includes usernames, email addresses, and perhaps sensitive academic information. To avoid abuse, data breaches, and unauthorised access, personal data must be maintained securely. Techniques for secure transmission protocols, like HTTPS, are crucial for safeguarding user data.

GDPR Compliance:

All organisations processing personal data of persons inside the European Union (EU) are subject to the General Data Protection Regulation (GDPR), a comprehensive data protection regulation. The Q&A platform is a prototype, but if it handles any personal data belonging to EU citizens, it still needs to abide by GDPR regulations.

Important GDPR tenets that apply to the Q&A Portal platform are as follows:

Fairness, Transparency, and Lawfulness: Processing of personal data must be done so in a way that is transparent, fair, and lawful. Users' consent must be sought when required, and they must be informed of how their data will be used.

Data minimisation: Only information required to operate the platform should be gathered. For user management, for instance, just the information required, such usernames and email addresses, is gathered.

Limitation on use: Personal information can only be gathered for explicit, transparent, and legitimate objectives. This implies that, in the case of a Q&A site, user data is only utilised for the functions that the platform intends, including publishing and account management.

Data accuracy: The platform has to make sure that user information is current and correct. To ensure accuracy, users must be able to update their personal data.

Storage restrictions: No personal information should be kept around for longer than is required. This entails putting in place a data retention policy for a Q&A site that specifies exactly how long user data is kept on file and when it will be removed.

Integrity and secrecy: Appropriate security measures must be taken while handling personal data. This includes using encryption, access restrictions, and routine security audits to safeguard data from unauthorised or unlawful processing, accidental loss, destruction, or damage.

Accountability: Platform owners need to be able to show that they are adhering to the principles of the General Data Protection Regulation (GDPR). This includes keeping track of all data processing operations and being ready to reply to enquiries from data subjects regarding things like data access or deletion.

Taking everything into consideration, In order to guarantee that the Q&A site is user-friendly, safe, and complies with data protection laws, it is imperative that legal, social, and ethical concerns be taken into consideration during the construction process. Ensuring a responsible and ethical approach to storing personal data in an educational context, the platform not only fulfils its legal duties but also builds trust and confidence among its users by incorporating robust online accessibility features and adhering to GDPR standards.

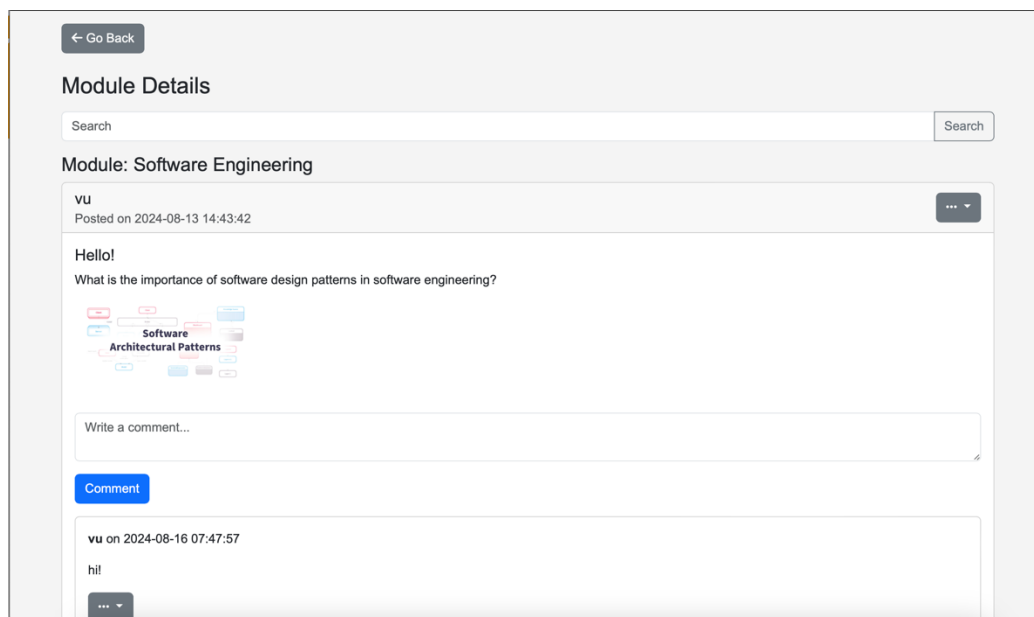
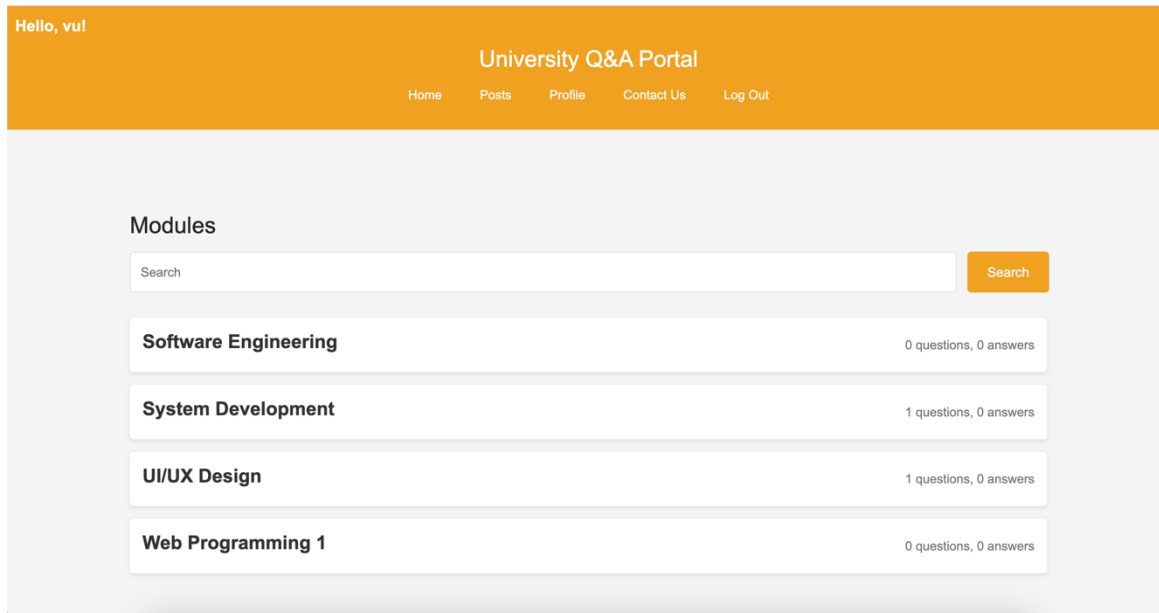
5. System Overview

The system successfully meets the following requirements:

Users will have the ability to add, amend, and remove their own postings from a list of queries or topics on the system. For added context, each post can have a picture or screenshot attached. In order to make sure that their postings are correctly connected in the database, users may also be maintained inside the system. This includes the option to modify or remove user details such as username and email address. Posts can also be associated with a module name that is chosen from an existing list and can be changed and handled as needed. Last but not least, a web form will enable users to email the administrator.

Function Of user:

Home Page: Users can view all topics on this site and there will be questions related to each topic. All related questions and answers are displayed. Users can edit, delete comments on their posts.



Post Page: Allows users to create a question form with or without an image corresponding to their desired topic.

The screenshot shows the 'Create a New Post' form in the University Q&A Portal. The form is titled 'Create a New Post' and includes the following fields:

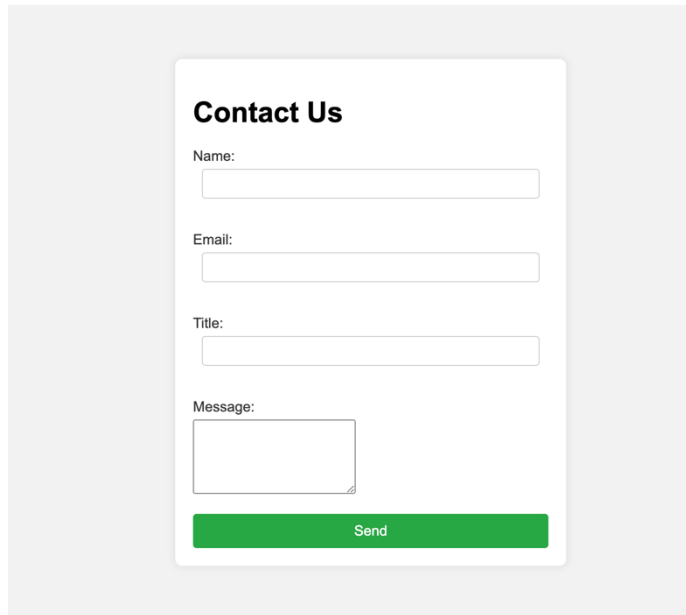
- Title:** A text input field containing 'Hello!'. Below the input is a small 'x' icon to clear the text.
- Content:** A large text area containing the text 'What is the importance of software design patterns in software engineering?'. A small 'x' icon is visible at the bottom right of the text area.
- Module Name:** A dropdown menu with 'Software Engineering' selected. A small 'v' icon is visible at the bottom right of the dropdown.
- Upload Image:** A section with a 'Choose File' button and a file name '1.jpg'.
- Add Post:** An orange button at the bottom of the form.

Users page: here allows users to change their email login password.

The screenshot shows the 'Update Profile' form in the University Q&A Portal. The form is titled 'Update Profile' and includes the following fields:

- Username:** A text input field containing 'vu'.
- Email:** A text input field containing '12@gmail.com'.
- Current Password:** A text input field.
- New Password:** A text input field.
- Confirm New Password:** A text input field.
- Update Profile:** A blue button at the bottom of the form.

Contact form: allows users to send requests or feedback to the admin via email.



The image shows a contact form titled "Contact Us". It is a white rectangular box with rounded corners, centered on a light gray background. The form contains four input fields: "Name:", "Email:", "Title:", and "Message:". Each field is a simple white rectangle with a thin gray border. Below the "Message:" field is a green button with the word "Send" in white text. The "Message:" field is a larger rectangle with a small icon in the bottom right corner, indicating it might be a text area or have a placeholder image.

Function Of admin:

Home Page: In terms of admin, this will be the page where all comments and posts are maintained, along with information about the poster, module_id, text of the questions and answers, images, and posting time. It will also include a section similar to a user function where administrators may ask enquiries.

Hello, phuong!

University Q&A Portal

[Home](#) [Manage User](#) [Subjects](#) [Log Out](#)

Admin Dashboard

Manage Posts

ID	User	Module	Title	Content	Image	Actions
135	phuong	UI/UX Design	123	123		
134	vuphuong	Software Engineering	hi!	how to create many databases?		
129	vu	Web Programming 1	Hello!	What are the key differences between server-side and client-side programming?	No Image	
127	vu	UI/UX Design	Hello!	What are the key differences between UI (User Interface) and UX (User Experience) design?		
125	vu	Software Engineering	Hello!	What is the importance of software design patterns in software engineering?		

Manage Replies

ID	Post	User	Content	Actions
70	Hello!	vu	Server-side processes have access to the server's resources, such as its CPU, memory, and storage, as well as any databases or other servers that the web application uses. Client-side processes, on the other hand, have access only to the resources of the user's device, such as its CPU, memory, and storage.	

Create a New Post

Title:

Content:

Please fill out this field.

Module Name:

Upload Image:

Manage User Page : An administrator can change, add, or remove personal information. They can also switch between being a user and an admin.

Hello, phuong!

University Q&A Portal

[Home](#)[Manage User](#)[Subjects](#)[Log Out](#)

Add New User

Username

Email

Password

Confirm Password

Add User

Existing Users

Email

Password

Confirm Password

Add User

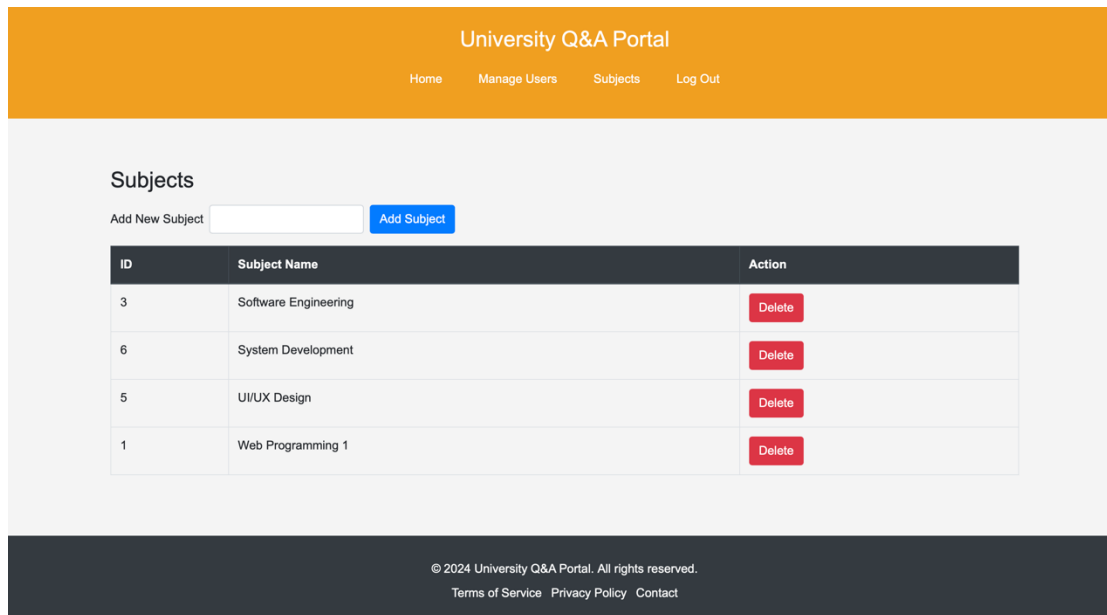
Existing Users

Username	Email	Created At	Action
phuong	345@gmail.com	2024-08-12 09:20:47	EditDelete
vu	12@gmail.com	2024-08-12 09:33:07	EditDelete
zu	123456@gmail.com	2024-08-19 09:05:40	EditDelete

© 2024 University Q&A Portal. All rights reserved.

Terms of Service

Manage Module Page: gives administrators the ability to control the names of newly added or removed modules.



6. Testing and Validation

The main features of the web-based CRUD system are described in the test table that follows, with an emphasis on user interactions, CRUD operations, role-based access control, security precautions, and responsive design. Every test is made to make sure the system functions as it should in different situations, confirming the platform's security and user experience.

Test ID	Feature	Test Description	Test Data	Expected Result	Pass/Fail
01	User Registration	Test the user registration process	Valid user details	The user should be registered, and a confirmation message should appear. The user should be able to log in.	Pass
02	User Login	Test the user login process	Valid username and password	The user should be logged in and redirected to the Home Page.	Pass
03	CRUD Operations - Create Post	Test creating a new post as a logged-in user	Valid post title and content	The new post should be created and displayed on the module page.	Pass
04	CRUD Operations - Edit Post	Test editing a post as the post owner and as an admin	1. Click the 'Edit' button on a post. 2. Modify and save changes.	The post should be updated with the new content.	Pass
05	CRUD Operations - Delete Post	Test deleting a post as the post owner and as an admin	1. Click the 'Delete' button on a post. 2. Confirm deletion.	The post should be removed from the module page.	Pass
06	Commenting	Test adding a comment to a post	Valid comment data	The comment should be displayed under the post.	Pass
07	Image Upload	Test image upload functionality	Valid image file	The image should be displayed	Pass

		during post creation		alongside the post.	
08	Role-Based Access Control	Test access control for editing and deleting posts and replies for different roles	Valid role actions (create, edit, delete)	Users should only be able to edit/delete their own posts. Admin should be able to edit/delete any post.	Pass
09	Navigation Menu	Test that the correct navigation items are displayed based on user role	1. Log in as a regular user. 2. Check the navigation menu. 3. Log in as an admin and check the menu.	The regular user should see 'Profile'; the admin should see 'Manage' and 'Subjects'.	Pass
10	Form Validation	Test form validation for registration, login, and post creation forms	1. Navigate to the relevant form pages. 2. Submit empty or incorrect data (e.g., invalid email).	The form should not submit, and appropriate error messages should appear.	Pass
11	Search Functionality	Test the search functionality within a module	1. Enter a search term in the search bar. 2. Submit the search.	Relevant posts matching the search term should be displayed.	Fail
12	Security - SQL Injection	Test protection against SQL injection	1. Attempt to inject SQL commands into input fields.	The system should sanitize inputs and prevent SQL injection.	Fail
13	Security - XSS	Test protection against Cross-Site	1. Attempt to inject malicious scripts into text fields.	The system should escape outputs and prevent XSS attacks.	Fail

		Scripting (XSS)			
14	Responsive Design	Test the layout and usability of the application on different screen sizes (mobile, tablet, desktop)	1. Open the application on different devices or use browser developer tools. 2. Check the layout and usability.	The application should be fully usable and visually appealing across all tested screen sizes.	Fail

Taking everything into consideration, The main features, such as user registration, login, CRUD operations, and role-based access control, are operating as expected, according to the tests that were carried out. But more work has to be done to fix problems with search functionality, security flaws like SQL injection and XSS, and making sure the design is responsive on all platforms. It will be essential to address these problems in order to provide a safe and intuitive system.

7. Evaluation, Conclusion, and Future Recommendations

Upon system review, the platform was found to be well-developed and to include numerous essential features, including user registration, login, and CRUD operations. These features were found to perform well and meet the project's essential needs. The effective integration of these functionalities guarantees that users may engage with the system meaningfully, effortlessly adding, modifying, and controlling material. As planned, role-based access control enabled administrators to oversee all postings and responses while limiting user access to only the material they created. This system's fundamental component creates a solid basis for a dependable and expandable platform.

Nevertheless, the testing stage brought to light a number of issues that need to be addressed right now. Although there was search capabilities inside the modules, it did not yield relevant results, which might potentially negatively impact the user experience by making it more difficult for users to locate certain material. In addition, serious security issues were found, especially with relation to vulnerabilities related to SQL injection and Cross-Site Scripting (XSS). The integrity of the system is seriously threatened by these problems, which might be used by hostile users to get unauthorised access or insert malicious scripts. Furthermore, it was discovered that the application's responsive design was deficient, since the UI was unable to adjust to various screen

sizes and devices with ease. This may have a detrimental effect on user accessibility, particularly in light of the growing use of mobile devices for internet access.

Although the system is well-founded, in order to guarantee user happiness and safeguard against any threats, vulnerabilities and design defects must be addressed. Enhancing responsive design for usability across all devices, putting strong security measures against SQL injection and XSS attacks in place, and improving search capabilities for effective content retrieval are priorities. Incorporating regular security audits and usability testing into the development process might help identify such problems early on. The present system will be strengthened by these enhancements, which will offer a safe, user-friendly experience that satisfies user requirements and industry standards. They will also establish a strong basis for future system expansions and upgrades.

Reference:

- W3C. "Web Content Accessibility Guidelines (WCAG) 2.1." Available at: <https://www.w3.org/TR/WCAG21/>
- U.S. Department of Justice. "Americans with Disabilities Act (ADA)." Available at: <https://www.ada.gov/>
- UK Government. "Equality Act 2010." Available at: <https://www.legislation.gov.uk/ukpga/2010/15/contents>
- Information Commissioner's Office (ICO). "Guide to the General Data Protection Regulation (GDPR)." Available at: <https://ico.org.uk/for-organisations/guide-to-data-protection/guide-to-the-general-data-protection-regulation-gdpr/>
- European Commission. "EU GDPR." Available at: https://ec.europa.eu/info/law/law-topic/data-protection/eu-data-protection-rules_en

Appendix:

This is a link of source code (gg drive):

<https://drive.google.com/drive/folders/1aZybYBcmD7eU9RdMPRBeGj2GtufnHQif?usp=sharing>

This is a link of video demonstration (gg drive):

<https://drive.google.com/file/d/1BKT9CvJB0tNC8JI2qqsdIP4e2lvAjCQE/view?usp=sharing>